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Artificial Intelligence and its Current Role in Clinical Outcome Prediction, Musculoskeletal Imaging, and Economic and Ethical Considerations within Orthopedics and Sports Medicine.

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CLINICAL & AGENTIC AI TAKEAWAY

AI models show promise in predicting patient outcomes, surgical complications, and healthcare utilization in orthopedics and sports medicine, but challenges in validation, accessibility, and ethical considerations remain.

ABSTRACT

PURPOSE OF REVIEW: Artificial intelligence (AI) has emerged as a useful tool across the field of orthopedic surgery. This review highlights recent literature on AI's role in surgical outcome prediction, musculoskeletal imaging, economic and ethical considerations, with a focus on its integration in sports medicine workflow and procedures. RECENT FINDINGS: Machine learning AI models have demonstrated superior accuracy in predicting orthopedic related patient-reported outcomes, surgical complications, and the utilization of healthcare compared to traditional, non-AI methods. Within imaging, AI applications now produce automated measurements for clinical and presurgical planning with precision equivalent to expert-level measurements. Large language AI models are increasingly used for clinical documentation, research workflows, and administrative support for healthcare delivery and effectiveness. Despite increasing integration of AI into orthopedics and its subspecialties, challenges in validation, accessibility due to cost, and ethical considerations remain. Orthopedic surgery and sports medicine are particularly well suited for AI applications due to their well-defined, measurable clinical outcomes. Emerging AI tools and models show promise in enhancing patient outcomes, surgical planning, and healthcare efficiency. Continued AI research must prioritize external validation, ethical implementation, and educational integration to ensure responsible, effective, and reproducible use.

Keywords: Artificial intelligence, Imaging, Sports Medicine, Orthopedic Surgery, Machine Learning, Clinical Outcome Prediction

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Abstract

Purpose of Review

Artificial intelligence (AI) has emerged as a useful tool across the field of orthopedic surgery. This review highlights recent literature on AI's role in surgical outcome prediction, musculoskeletal imaging, economic and ethical considerations, with a focus on its integration in sports medicine workflow and procedures.

Recent Findings

Machine learning AI models have demonstrated superior accuracy in predicting orthopedic related patient-reported outcomes, surgical complications, and the utilization of healthcare compared to traditional, non-AI methods. Within imaging, AI applications now produce automated measurements for clinical and presurgical planning with precision equivalent to expert-level measurements. Large language AI models are increasingly used for clinical documentation, research workflows, and administrative support for healthcare delivery and effectiveness. Despite increasing integration of AI into orthopedics and its subspecialties, challenges in validation, accessibility due to cost, and ethical considerations remain.

Summary

Orthopedic surgery and sports medicine are particularly well suited for AI applications due to their well-defined, measurable clinical outcomes. Emerging AI tools and models show promise in enhancing patient outcomes, surgical planning, and healthcare efficiency. Continued AI research must prioritize external validation, ethical implementation, and educational integration to ensure responsible, effective, and reproducible use.

Keywords:Artificial intelligence, Machine learning, Orthopedic surgery, Clinical outcome prediction, Imaging, Sports medicine

Introduction

Orthopedic surgery is undergoing a technological revolution driven by the integration of artificial intelligence (AI) into diagnostic, surgical, and rehabilitative processes. AI has become a tool capable of enhancing clinical decision making, streamlining workflow efficiency, and personalizing patient care. This transformation parallels the broader digital

evolution of orthopedic surgery, as data-driven approaches and robotics are increasingly guiding clinical decision-making and patient outcomes [1–3].

Clinical and preoperative care are among the most active domains for AI integration, with demonstrated success in imaging interpretation and automation [4,5]. For example, in anterior cruciate ligament (ACL) reconstruction, deep learning (DL) AI models can quantify posterior tibial slope from radiographs with surgeon-level accuracy, enabling anatomic risk assessment for the potential risks of graft failure [6]. Similarly, machine learning (ML) AI models have enhanced prediction of ACL reconstruction revision risk based on patient specificities supporting individualized preoperative planning [7].

As AI adoption expands within orthopedic surgery, careful evaluation of its capabilities and limitations is essential. The following subsections summarize core applications of AI across orthopedics and the subspeciality of sports medicine by providing in-depth discussion of outcome prediction in upper and lower extremity procedures, imaging automation and quantitative analysis, as well as economic and ethical considerations. This current concept review aims to provide a broad synthesis of contemporary applications and future directions for the clinically meaningful implementation of AI in musculoskeletal care.

Core Applications of AI in Sports Medicine

Outcome Prediction and Patient Selection

AI outcome prediction has been applied across upper and lower extremity sports medicine procedures, enabling joint-specific risk stratification and functional outcome prioritization. ML algorithms trained on clinical and imaging datasets enable predictive models to estimate a patient's likelihood of achieving functional milestones or experiencing surgical complications in individualized and selected cohorts. These models are particularly valuable for high-volume procedures such as rotator cuff repair, where identifying patients at risk for suboptimal recovery can enhance preoperative counseling and postoperative planning [8,9]. Efforts to standardize AI applications in outcome prediction underscore the need for transparent reporting of AI model training, comprehensive dataset composition, and AI performance metrics to ensure meaningful clinical utility [10].

Recent AI models enable quantification of individual clinical contributors to injury risk. In professional baseball athletes, these models demonstrated strong predictive performance (AUC > 0.80), with prior injury history and workload-related performance metrics emerging as the dominant predictors compared with traditional single-variable analyses (Fig.1) [11]. Such insights reinforce the validity of AI outputs and support their use in prehabilitation planning, workload monitoring, and risk stratification.

Variable importance plot for predicting injury risk in professional athletes. A machine learning (ML) model's assessment of the most influential factors contributing to future injury risk in Major League Baseball players. Prior injury history and advanced performance metrics, such as weighted cutter runs per 100 pitches and wins above replacement, ranked highest in predictive value. Visualizations like this enhance model interpretability and support the integration of artificial intelligence (AI)-based tools into personalized prehabilitation strategies and athlete risk profiling.

Adapted from Karnuta et al.[11]

In the lower extremity, Kunze et al. validated ML algorithms capable of predicting clinically meaningful improvement following hip arthroscopy for femoracetabular impingement (FAI) [12,13]. The algorithms incorporated preoperative variables such as demographics, baseline clinical scores, and interoperative findings to generate individualized outcome estimates [14,15].

Subsequent external validation confirmed the model's reliability, reinforcing that well-designed algorithms can retain predictive accuracy across institutions and patient demographics [15]. ML models demonstrated fair to good discrimination for predicting subsequent hip procedures, including need for revision arthroscopy and progression to total hip arthroplasty (Fig.2). Model performance was strongest for predicting progression to total hip arthroplasty, with an area under the curve (AUC) of approximately 0.80, and demonstrated moderate discrimination for revision arthroscopy and periacetabular osteotomy (AUCs ≈ 0.75–0.77) [16]. This reproducibility is critical for advancing AI predictive tools from single-center studies to large-scale clinical use, where variability in patient populations and practice settings can otherwise compromise validity and reliability.

Receiver operating characteristic (ROC) curves for predicting subsequent hip surgeries using machine learning (ML). ROC curves display the predictive performance of random forest models developed to forecast the risk of subsequent hip procedures following primary hip arthroscopy for femoroacetabular impingement (FAI). Area under the curve (AUC) values were 0.77 for revision hip arthroscopy, 0.80 for total hip arthroplasty (THA), 0.62 for hip resurfacing arthroplasty (HRA), and 0.76 for periacetabular osteotomy (PAO), reflecting fair to good discriminative accuracy. These models integrate preoperative demographic, radiographic, and functional data to assist in surgical risk stratification and patient counseling

Despite this progress, widespread clinical translation has remained limited. Most models demonstrate strong internal validation but lack external validation across diverse populations and practice settings [17]. The absence of replicability and widespread access constrains their universal utility. Integration into clinical workflows will require emphasis on interpretability and evaluation of bias among clinicians. In sports medicine, where return-to-sport (RTS) and long-term functional success define treatment goals, AI prediction platforms have the potential to become staple components of individualized surgical and non-surgical treatment. As healthcare increasingly transitions towards value-based care, AI-based tools capable of estimating patient satisfaction, functional recovery, and treatment efficacy will become central to optimizing patient care delivery [18]. These upper and lower extremity applications highlight AI's capacity to enhance care from outcome prediction to improved patient experience through more accurate patient assessment and treatment. As models continue to evolve, their integration into clinical pathways will expand, resulting in a new era of data-driven, patient-centered care.

Imaging and Biometric Analysis

AI is transforming the interpretation of orthopedic imaging. DL algorithms now demonstrate high accuracy in recognizing radiographic parameters and alignment abnormalities [19]. In ligamentous knee injuries, ML can support more informed decision making by standardizing the assessment of imaging-derived risk factors known to influence graft failure risk [7]. Tasks such as posterior tibial slope quantification, once prone to interobserver variability, can now be standardized through automated algorithms [6]. In patients with ACL injury, DL models have demonstrated close agreement with expert human measurements, achieving a mean absolute error of approximately 2° while reducing processing time to approximately 5 seconds per image compared with an estimated 180 minutes for manual segmentation and measurement [6].

The utility of AI also extends to subtle injury detection. In hip preservation surgery, radiographic indices of native anatomy and pathology derived from computed tomography (CT) can be paired with random forest classifiers to predict long-term functional outcomes, illustrating a typical imaging-driven approach to outcome prediction in hip arthroscopy (Fig.3). These methods underscore AI's impact on musculoskeletal radiology, particularly in high volume or resource-limited environments where automation can reduce time and error.

Radiographic indices and ML pipeline for predicting hip arthroscopy outcomes. This figure outlines the ML workflow used to evaluate the prognostic value of preoperative CT-based radiographic indices in patients undergoing hip arthroscopy. Radiologic angles were extracted by board-certified radiologists and analyzed using a random forest classifier to predict one- and two-year patient-reported outcomes, including Modified Harris Hip Score (mHHS), Hip Outcome

Score – Sport Specific (HOS-SS), Hip Outcome Score - Activities of Daily Living (HOS-ADL), and International Hip Outcome Tool (iHOT-33). The model's performance was assessed using the AUC, as shown for both timepoints.

Adapted from Ramkumar et al.[14]

However, as reliance on algorithmic outputs increases, concerns are warranted regarding clinical oversight and algorithmic bias, particularly when models trained on limited or non-representative datasets are applied across diverse patient populations [20]. Optimizing widespread deployment will require standardized and protocol-driven approaches for integrating multimodal imaging inputs including magnetic resonance imaging (MRI), CT, and radiographs into unified predictive models [21,22].

For example, a study by Ramkumar et al. used a ML model to predict functional recovery after hip arthroscopy by incorporating imaging and radiographic morphology as potential predictors. The cohort reflected a wide range of morphology across femoral and acetabular parameters, including alpha angle, center-edge angles, version profiles, and combined indices such as the McKibbin and hip impingement indices (Table1). Tables2and3demonstrate that isolated radiographic indices show limited utility for functional recovery at 1–2 years [14]. Across sport-specific functions and activities of daily living, odds ratios remain close to null with non-significant associations, and discrimination is consistently modest, with AUC values clustering near 0.50–0.57 across most parameters [14]. Specifically, alpha angle and coronal center-edge angle both demonstrated odds ratios near 1.0 and AUCs near 0.50–0.55 at one- and two-year follow-up [14].

Baseline patient characteristics for machine learning (ML) cohort undergoing hip arthroscopy. This table outlines demographic, laterality, and radiographic characteristics for 1,735 patients who underwent hip arthroscopy. Radiographic angles include alpha, beta, neck shaft, acetabular version, and femoral version. These data were used as features in ML models predicting post-operative outcomes.

Adapted from Ramkumar et al.[14]

αData are shown as mean ± standard deviation unless otherwise indicated

Change in sport function (HOS-SS) by radiographic subgroup. Mean improvement in HOS-ADL scores is displayed across various anatomical groupings (e.g., alpha angle ≥ 55° vs. < 55°). No statistically significant differences were observed, suggesting radiographic measures do not strongly correlate with recovery in daily activities.

Adapted from Ramkumar et al.[14]

αAUC, area under the curve; CI, confidence interval; HOS-SS, Hip Outcomes Score-Sport Specific

Change in daily living function (HOS-ADL) by radiographic subgroup. This table presents mean changes in HOS-S scores following hip arthroscopy, stratified by radiographic parameters. As with HOS-ADL, minimal variation between groups highlights the limited prognostic value of static anatomical indices.

Adapted from Ramkumar et al.[14]

αAUC, area under the curve; CI, confidence interval; HOS-ADL, hip outcomes score-sport specific

These findings suggest that commonly cited anatomic indices alone do not meaningfully stratify postoperative functional improvement and should not be used as sole inputs for patient selection or counseling. Instead, they support a more clinically useful paradigm where radiographic morphology is used within multivariable models that incorporate baseline function, intraoperative findings, and patient-level factors of recovery. This shifts AI-driven patient selection from morphology-only prediction on imaging to multidimensional prognostication [14].

Machine Learning and Time- and Cost-Effective Care Delivery

AI also offers a powerful means to enhance value-based care through cost prediction and improved operational efficiency. ML models consistently outperform traditional heuristics and clinician judgment in forecasting surgical costs, discharge disposition, and inpatient resource utilization [9]. These predictive tools can identify high-cost drivers preoperatively. This allows providers and payers to anticipate factors such as prolonged operative time, inpatient admission, or post-acute care needs and allocate resources accordingly [9]. A simplified decision tree (Fig.4) illustrates how individual clinical inputs are processed to generate a cost estimate.

Decision tree example for predicting cost following anterior cruciate ligament (ACL) reconstruction

A decision tree used by a ML model to predict average total charges for ACL reconstruction based on key perioperative variables. The model first stratifies cases by anesthesia type and operative time, with green nodes representing lower-cost predictions. Such tools can facilitate real-time decision-making and cost forecasting in surgical planning.Adapted from Lu et al.[23].

Models predicting clinically meaningful improvement also support value-driven care pathways. For example, the Orthopedic Sports Medicine and Shoulder Outcomes (OSSO) algorithm predicted attainment of the minimal clinically important difference (MCID) following hip arthroscopy with similar performance in the development and external validation cohorts, with c-statistics of 0.77 and 0.80, respectively (Table4). Such reproducibility enhances clinical confidence and supports health system-wide implementation for consistent outcome tracking.

Comparative performance of the OSSO machine learning (ML) algorithm in development and external validation cohorts. This table presents a side-by-side comparison of performance metrics for the OSSO minimal clinically important difference (MCID) prediction model in patients undergoing hip arthroscopy. The model demonstrated strong generalizability, with similar c-statistics, calibration slopes, and Brier scores across both the original internal validation and the external validation cohorts. Decision curve analysis revealed greater net clinical benefit compared with other management strategies.

Adapted from Kunze et al.[12]

Values are presented as mean (95% confidence interval). c-statistic, concordance statistic; ML, machine learning; OSSO, Orthopedic Sports Medicine and Shoulder Outcomes

Procedure-specific economic modeling has likewise gained traction. AI-based frameworks can integrate clinical, radiographic, and demographic data to predict the financial burden of surgeries such as ACL reconstruction [23]. This capability is particularly relevant as bundled payments and outcome-linked reimbursement models become standard across the practices of orthopedic surgery.

Emerging Technologies and Large Language Models

Large language models (LLMs) have recently emerged as tools capable of advancing musculoskeletal workflow. These systems, such as chat-based LLMs, can draft clinical notes, simplify medical instructions, and clarify highly complex information into patient-accessible language, thereby improving patient adherence and satisfaction [24–26]. However, while LLMs offer a promise of streamlined communication, recent reviews indicate that their performance remains inconsistent. Issues such as hallucinated responses, lack of source transparency, and poor reproducibility raise concerns about reliability in clinical environments [27–29]. As a result, an integral component to ensure their safe and effective utilization is developing solid study designs, openly using the training data, and providing domain-specific fine-tuning approaches.

By processing natural language inputs at scale, these systems can automate complex tasks such as literature review and synthesis, and protocol drafting. Different LLM model types have diverse input–output modalities and functional

capabilities. Appropriate alignment between model capabilities and intended use may help reduce clinician burden and improve consistency in data-driven care pathways (Table5) [29]. In practice, inadequate alignment between model capability and intended use has contributed to variable methodological quality in the literature, with systematic reviews identifying redundancy, limited rigor, and insufficient customization across healthcare applications [28]. These findings emphasize the need for purposeful model design, transparent evaluation, and orthopedic specific modeling prior to clinical implementation.

Model concepts and functions of basic large language models (LLMs). Input-output modalities and core functions of LLM types, highlighting distinctions between task-based models and foundational systems.

Adapted from Kunze et al.[29]

3D, 3-dimensional; LLM, large language model

Effective use of LLMs in orthopedics depends not only on algorithmic performance, but also on clinician literacy and judgment. As AI platforms become more accessible, orthopedic surgeons must learn to interpret, validate and deploy output responsibly to ensure alignment with the most responsible patient care. Educational reform is recognized as essential to improving AI fluency among trainees and practitioners [25,30]. Retrieval augmented generation (RAG) models have demonstrated superior evidence synthesis compared with general purpose LLMs, underscoring the value of tailored customization [27].

In parallel, natural language processing (NLP) has become an important modality for extracting structured insights from unstructured clinical text. By mining electronic medical records (EMR), NLP algorithms can identify trends, stratify patient risk, and facilitate registry creations. These functions are otherwise limited by human bandwidth and variability [31]. The implications for research efficiency, quality assurance, and personalized care delivery are established, particularly as orthopedic institutions seek to build scalable data infrastructure.

When viewed collectively, the implementation of AI across imaging, prediction, and administration provides clear support for its general utility in orthopedics. Whether by improving diagnostic precision, optimizing resource allocation, or supporting communication, AI has the potential to drive a new era of efficient, personalized musculoskeletal care. In both academic and community healthcare settings, the collective impact of AI suggests a unifying benefit: higher-quality musculoskeletal care delivered more efficiently and at a lower cost.

Ethical, Educational, and Global Considerations

The integration of AI into orthopedic surgery introduces complex ethical, educational, and equity-based challenges that need to be addressed to ensure responsible implementation. One of the central concerns is premature clinical adoption of unvalidated AI tools, a phenomenon termed the “valley of despair”, where enthusiasm surpasses its actionable benefit [32]. To mitigate this, bioethical frameworks are being developed to guide AI deployment with principles centered on transparency, equity, and safety [33,34]. These frameworks advocate for complete model testing, clear documentation of limitations, and mechanisms to ensure human oversight remains central to the decision-making process.

From an educational standpoint, there is growing recognition that AI competency should be integrated into orthopedic training at every level. Recent surveys of orthopedic trainees reveal significant variability in AI knowledge, highlighting the need for structured exposure to these technologies early in surgical education [35]. Without a foundational understanding, clinicians risk overreliance on or misinterpretation of AI-generated outputs. Editorial voices have underscored the danger of deploying AI tools without sufficient user competency, equating these actions to the introduction of unregulated instruments into the operating room [36–38]. Multimodal LLMs can process and summarize radiologic findings, a capability that underscores both their potential utility and the need for user training and oversight (Fig.5).

Multimodal large language model (LLM) for medical imaging interpretation. This diagram illustrates how a multimodal LLM integrates radiographic imaging with text-based prompts to generate clinical findings. A pretrained vision

transformer extracts features from the image (in this case, a chest radiograph), which are then processed through a linear transformation layer and interpreted by a medically trained LLM. The model outputs a structured diagnostic report, showcasing the potential of LLMs to enhance decision-making through image-text alignment.

Adapted from Kunze et al.[29]

Global equity remains another important consideration. Most existing AI models are trained and validated on data from high-income countries limiting their applicability in resource constrained regions risking digital health disparities. Ensuring fairness will require inclusion of diverse patient populations and open access international datasets to promote equitable outcomes [39,40].

Ultimately, the promise of AI in orthopedics depends not only on technological advancement, but also on adherence to ethical principles, comprehensive education, and global inclusivity. These pillars are essential for transforming AI from a theoretical innovation to a trusted partner in musculoskeletal care delivery.

Future Directions and Conclusion

AI continues to evolve in musculoskeletal medicine offering opportunities to enhance diagnostic accuracy, improve outcome prediction, and streamline care delivery and costs. As adoption progresses, the field must transition from proof-of-concept models to sustained clinical integration. The importance of robust model design, external validation, and ethical deployment across a broad range of orthopedic subspecialties has been clearly demonstrated by the contributions of physician-scientists thus far [24,29,31,41,42]. Ongoing translational scholarship will be essential in bridging the gap between data science and bedside application.

Despite significant progress, many existing algorithms remain limited by small training cohorts and inconsistent reporting standards. To achieve clinical utility, future efforts must prioritize interpretability and workflow compatibility, ensuring the AI tools enhance rather than complicate clinical decision making. As with any paradigm-shifting technology, enthusiasm should be balanced with critical appraisal and a commitment to patient centered care.

Priority areas for future development include:

Prospective, multicenter validation of ML and DL algorithms across diverse patient populations to ensure external validity and reduce algorithmic bias.

Standardized transparent reporting of model architecture and performance metrics, including calibration curves, confusion matrices, and AUC values, to support reproducibility and clinician confidence.

Development of formal orthopedic-specific educational initiatives and programs to improve AI fluency among trainees and practicing surgeons, ensuring responsible clinical use.

Integration of validated AI tools into real-time EMR systems and surgical planning platforms, facilitating decision support and reducing the cognitive burden of data management.

The next chapter of orthopedic innovation will depend on the ability to validate, educate, and integrate AI responsibly. If these priorities are met through multi-center collaboration, policy support, and academic rigor, AI will be well positioned to lead into a new era of orthopedics. This new era will be defined by clinical precision, personalization, and measurable improvements in healthcare and patient outcomes.

Key References

Author contributions

E.O. and B.S. wrote the main manuscript text. E.O. prepared all figures.All authors reviewed and edited the manuscript.

Data Availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

Disclosure statement

Statement: Below are the healthcare industry relationships reported by Dr. Nwachukwu as of July 8, 2025.

BICMD - Co-founder, Ownership Interest

Figur8 - Advisory Board

HSS West Side ASC - Ownership Interest

Medbridge- Advisory Board

Quantum - Advisory Board

Stryker Corporation - Consultant

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References

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This section collects any data citations, data availability statements, or supplementary materials included in this article.

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